Soundarya Pradhan

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EDUCATION

University of Illinois, Urbana Champaign - *Master of Computer Science*, GPA:3.83 Expected: May 2027
NIT, Jamshedpur - *Bachelor of Technology*, GPA: 7.85 2018-2022

SKILLS: Transformers, Unsloth, llama.cpp, Flair, Pytorch, Scikit-learn, Pandas, Numpy, App development, Screen Reader accessibility, User Research, Docker, wxPython, Tauri, C++, C, Git, MongoDB, SQL, Socket

PROFESSIONAL EXPERIENCE

Research Assistant, UIUC - *Adviser: Prof. JooYoung Seo* Jan 2026 - Present

- Contribute to MAIDR (open source TypeScript library) for accessible charts using sonification, braille, and text
- Built multi-line braille display support, letting users read several data series at once, with presets for 9+ commercial displays (e.g., Focus 40 Blue, Orbit Reader)
- Developed an algorithm to identify and navigate to line intersections and heatmap extrema, so users can find key data points without scanning the whole chart
- Designed pointer-based audio guidance using the Web Audio API that directs users toward chart elements, and improved stereo panning to convey horizontal position more smoothly

Skima Innovations - *ML Intern* Apr – Sep 2024

- Co-developed a multi-label encoder model to assign diversity and disability tags to resumes
- Converted a decoder model to bi-directional text encoder model, with continued pretraining on a corpus, ensuring its viability as backbone model for tasks such as NER, text embedding, and retrieval
- Finetuned a decoder model for NER, improving F1 by 0.9% compared to the existing encoder model

HSBC Electronic Data Processing, India - *Manager: Finance Systems Operations* July 2022 – Jun 2024

- Led the transition of front-end from tkinter to wxPython, ensuring 100% screen-reader accessibility
- Automated critical regulatory reporting (NSFR, Stress Testing, ALCM) using Python, improving data analytics efficiency by over 20%

PROJECTS

Distilling Mathematical Intelligence (STEM Accessibility Pipeline) 2025–2026

- Architected a two-phase curriculum distillation pipeline to convert complex STEM PDFs into accessible Markdown (Math to LaTeX, Diagram to Textual Description)
- Distilled a 27B teacher into a lightweight 4B student model achieving a 6.75x VRAM reduction

SK Chess 2025

- Pradhan, S. K., Anand, A., & Seo, J. (2026). SK Chess: An accessible chess system for Blind and Low Vision chess players. ACM ASSETS '26. https://doi.org/10.1145/3797867.3841238
- Created comprehensive accessible chess software enabling visually impaired users to analyze positions and prepare for games independently
- Features include specialized auditory notifications, backtracking assistance, and FIDE-compliant phonetic move pronunciation

Accessible Chess 2022

- Developed a website that converts chess positions from major software (Chessbase, Fritz) into human-readable text for visually challenged players
- Won the **NCPEDP-Mphasis Universal Design Awards** 2022
- Awarded a scholarship from Chessbase India in recognition of the project's high impact on the community